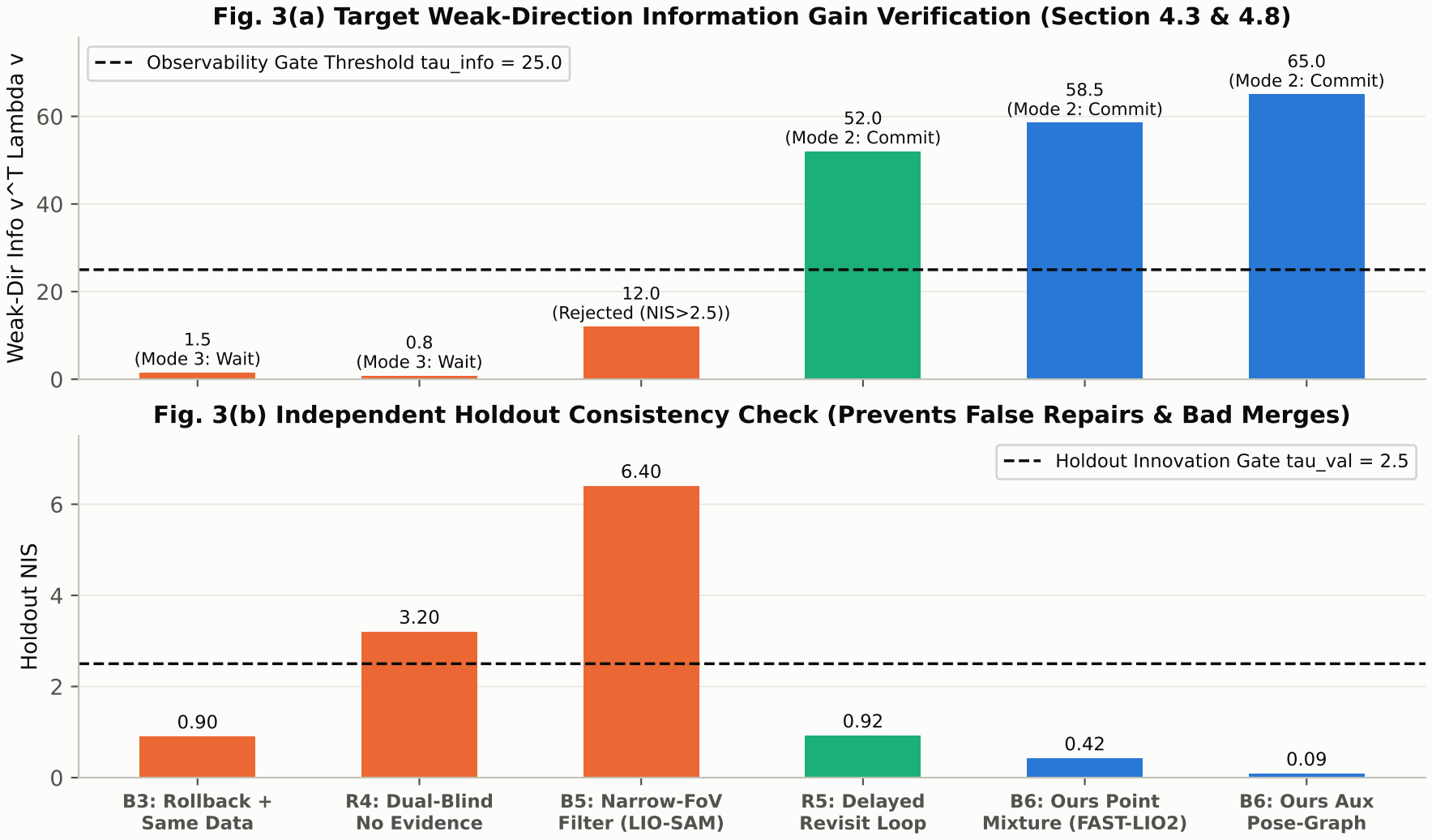
Fig. 3(a) Target Weak-Direction Information Gain Verification (Section 4.3 & 4.8)
Observability Gate Threshold tau_info = 25.0
Weak-Dir Info v^T Lambda v
0
20
40
60
1.5
(Mode 3: Wait)
0.8
(Mode 3: Wait)
12.0
(Rejected (NIS>2.5))
52.0
(Mode 2: Commit)
58.5
(Mode 2: Commit)
65.0
(Mode 2: Commit)
Fig. 3(b) Independent Holdout Consistency Check (Prevents False Repairs & Bad Merges)
Holdout Innovation Gate tau_val = 2.5
Holdout NIS
0
2
4
6
0.90
3.20
6.40
0.92
0.42
0.09
B3: Rollback + Same Data
R4: Dual-Blind No Evidence
B5: Narrow-FoV Filter (LIO-SAM)
R5: Delayed Revisit Loop
B6: Ours Point Mixture (FAST-LIO2)
B6: Ours Aux Pose-Graph